

Business Analysis with Unstructured Data: Airbnb Case Study

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Executive Summary

This report examines unstructured, textual and numerical data from Airbnb listings to develop valuable and actionable business insights. We employed text mining techniques - Sentiment Analysis, Topic Modeling , N-grams and Word Frequency Analysis - to investigate the unstructured description column and pull-out key business implications. Accordingly, we visually represented the data to tell a compelling story and to uncover actionable insights. In addition, we created an interactive Power BI and R-Shiny dashboard to fully represent these insights and allow the data to be displayed in a dynamic manner.

Business Insights

The analysis of sentiment indicated that the tone of Airbnb listing descriptions is mainly positive. Visualization of the most frequently occurring positive and negative words underscored the importance of cleanliness, comfort, and convenience in promoting guest satisfaction. Properties that included references to cleanliness, comfort, and convenience tended to have more positive responses, which suggests that hosts should advocate using that language in their property descriptions. Topic modeling (LDA) also revealed five themes that emerged frequently in listing; these themes were location, amenities, space, check-in experience, and attractions nearby. These data insights can be beneficial to Airbnb and its hosts in creating listings that align with the aspects that are most important to guests.

Other supporting analyses, including visualizing bigram frequencies, revealed repetition of phrases, such as "living room," "air conditioning," and "walking distance," that signify important qualities that attract guests. The word correlation models distinguishing types of rentals (Apartment, Condominium, and Serviced Apartment) demonstrated both commonality and variability in descriptors to help Airbnb understand how guests might feel differently about types of accommodations. Geographic observations from availability and price visualizations indicated high average prices and limited listings, especially in Hawaii, suggesting high demand and frequency of booking, among U.S.-based listings. This feedback can provide Airbnb data on improving the allocation of resources, marketing strategies, and recommendations to hosts on how to enhance an Airbnb listing to improve visibility and income.

Key Visualizations

The visualizations were plotted to investigate deep insights from the description column and other associated characteristics of Airbnb dataset. The “Most Common Positive and Negative Words” chart illustrated the general emotional tone in which hosts expressed themselves. One could see a clear typographical emphasis on positive terms, emphasizing how listings are marketed to prospective guests. The “Word Frequency Correlation: Apartment vs Others” focusing on similar terms used across property types, that is, apartments, condominiums and serviced apartments (each property type emphasizing the unique features of that property type). Some words were shared amongst all property types, demonstrating how certain terms are associated to different property types illustrating how language is manipulated to emphasize different value propositions. The “Average Price by State (US Only)” chart uncovered wide variance in pricing across properties and states (specifically pinpointing premium pricing in

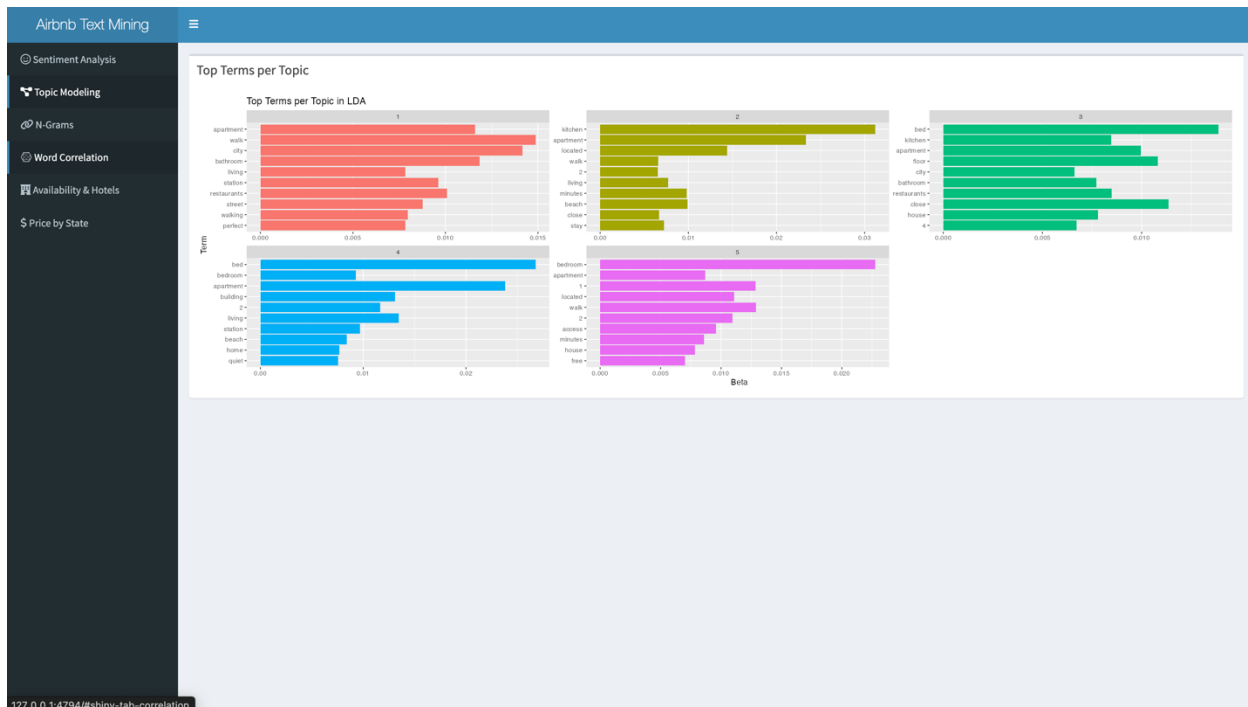
states associated with tourism). Lastly, the “Availability and Hotel Count Analysis by Country” analysis showed the USA is a hub for Airbnb listings and has a high order of listing availability and count, whereas countries with lower total listing counts (for example, China) demonstrating either lack of functionality of listings or regulatory controls in those areas.

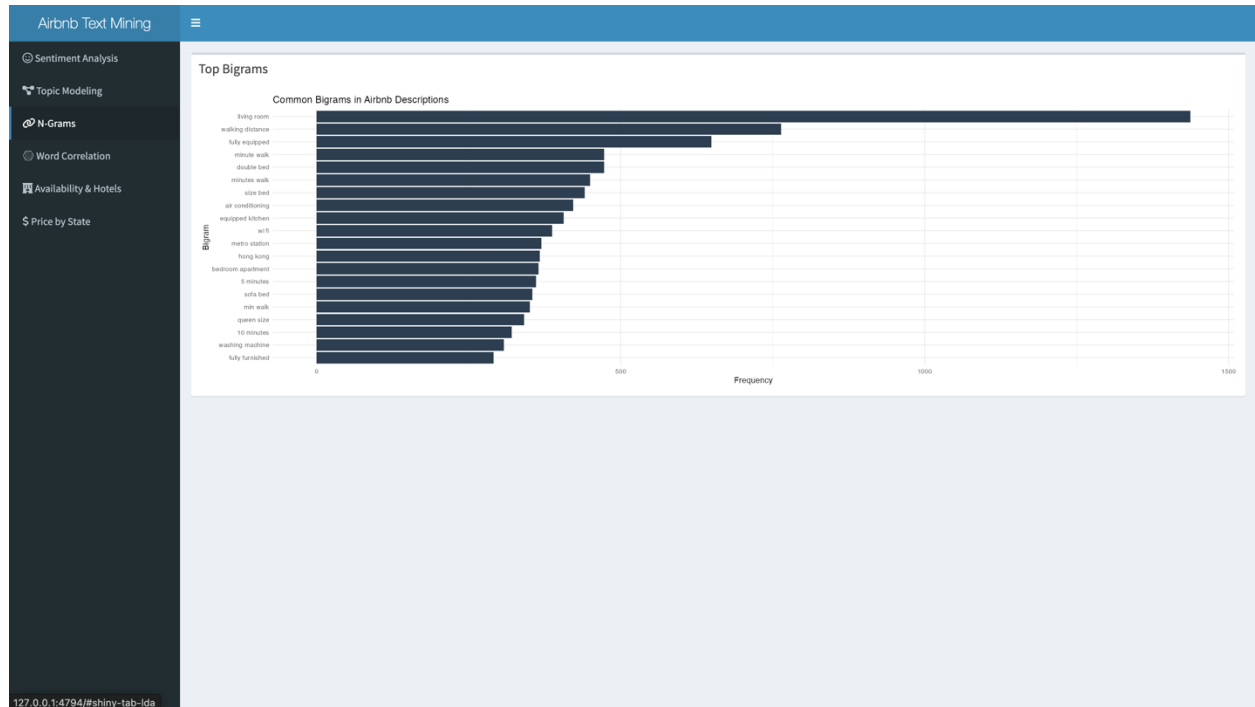
Dashboard Implementation

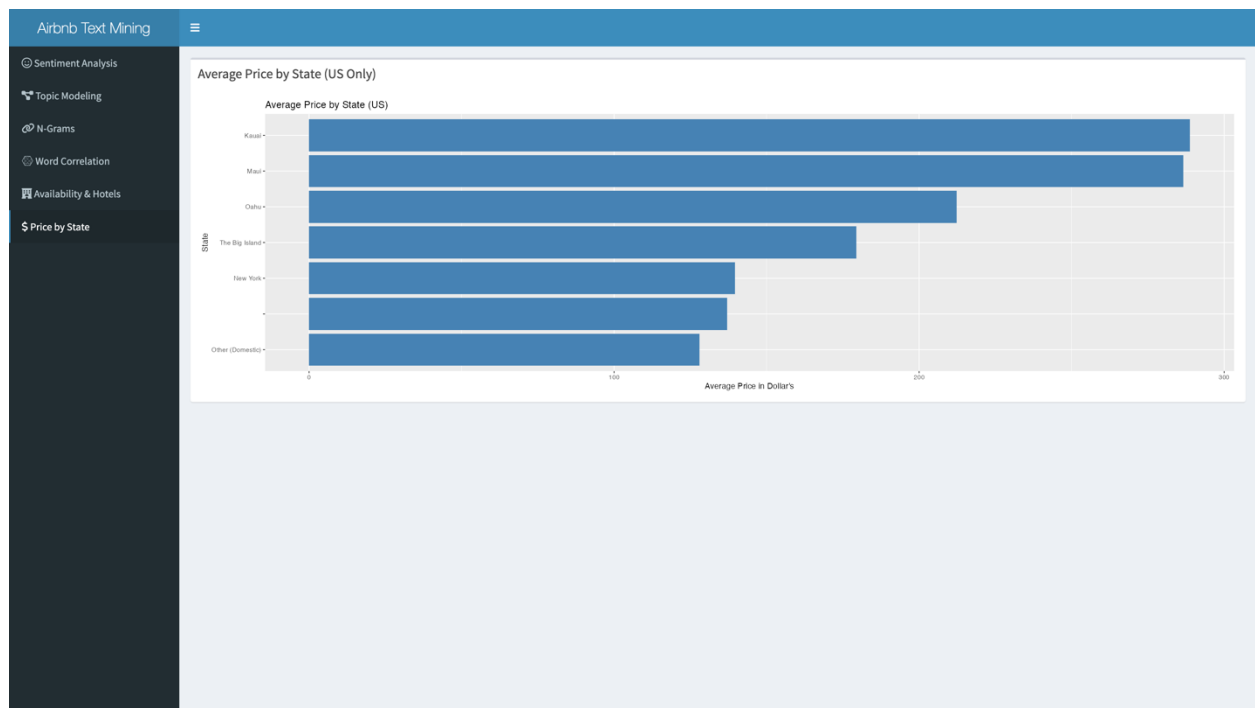
The dashboards implemented using R Shiny and Power BI provide a dynamic and interactive view of data related to Airbnb listings. The R Shiny dashboard allows modular navigation between several tabs titled Sentiment Analysis, Topic Modeling, N-Grams, and Word Correlation, which creates data informatics derived from textual data. Each tab contained clean visuals to represent the data on the dashboard. The Power BI dashboard allows users to view and examine both numerical data as well as sentiment data within the infographics simultaneously. Users can see KPIs such as total listings, average price, and availability all in one view of the dashboard. The Power BI dashboard also allows users, through a visual aid, to see the geographical data in illustrated form and indicates which sentiment words associated with a high frequency of use, thus allowing the reader to gain insight about the visibility and attraction of listings across countries and states.

Appendix

RSHINY DASHBOARD







POWER BI DASHBOARD

